ECOSENSE

Smart Energy Monitoring, Waste Detection & Saving Recommendation System

Rathinam Global Deemed to be University

Department of Physics · Team Kinetic Minds

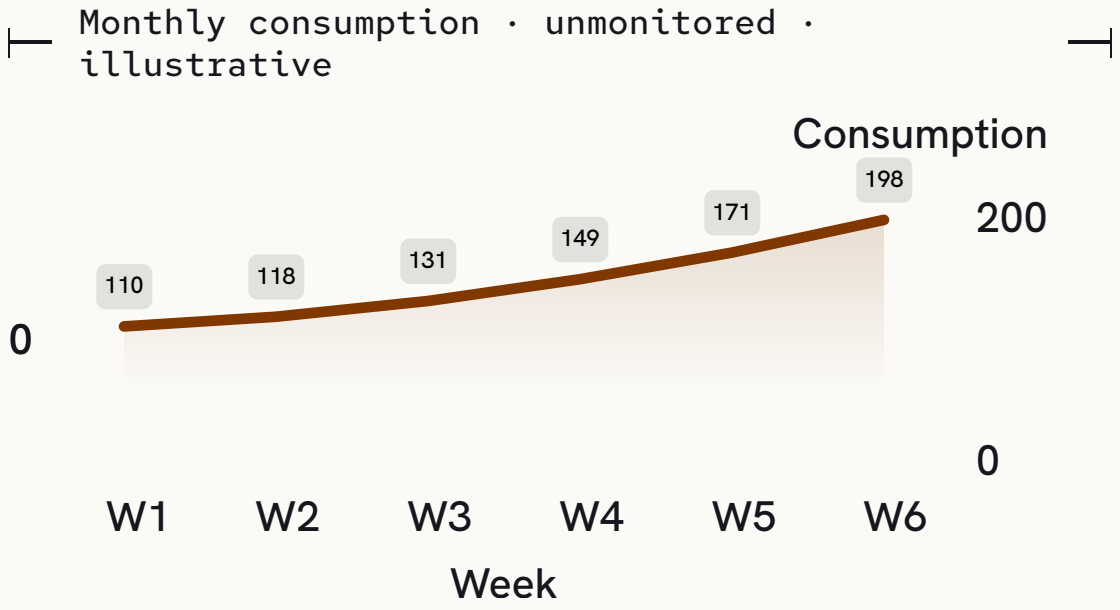
Measure · Analyze · Detect · Recommend · Save



THE PROBLEM: ENERGY WASTE IS OFTEN INVISIBLE

1. Real-time consumption is hard to see; unusual usage can go unnoticed.
2. Devices may run longer than necessary; excess is discovered only after the bill.
3. Little guidance on achievable savings.

How can we identify unnecessary energy usage and help users reduce it?



Without monitoring, rising consumption can remain hidden until the bill arrives.

ECOSENSE: FROM MONITORING TO ACTION

EcoSense measures consumption and analyzes usage patterns to identify potential energy waste.

Stage	System function
Monitor	Real-time energy monitoring
Analyze	Historical usage comparison
Detect waste	Unusual usage and Waste Score
Recommend	Saving recommendations
Save	What-if saving simulation

SYSTEM ARCHITECTURE

Electrical signal → decision support

Hardware

Electrical Load
Appliance

Current Sensor
Measures current

ESP32
Processes data

Power / Energy
Calculates P and kWh

Wi-Fi
Transmits data

Software

EcoSense Web App
Dashboard

Analysis
History, budget, operating time

Waste Detection
Waste Score

Recommendation
Action to reduce usage

Saving Simulation
What-if result

Power

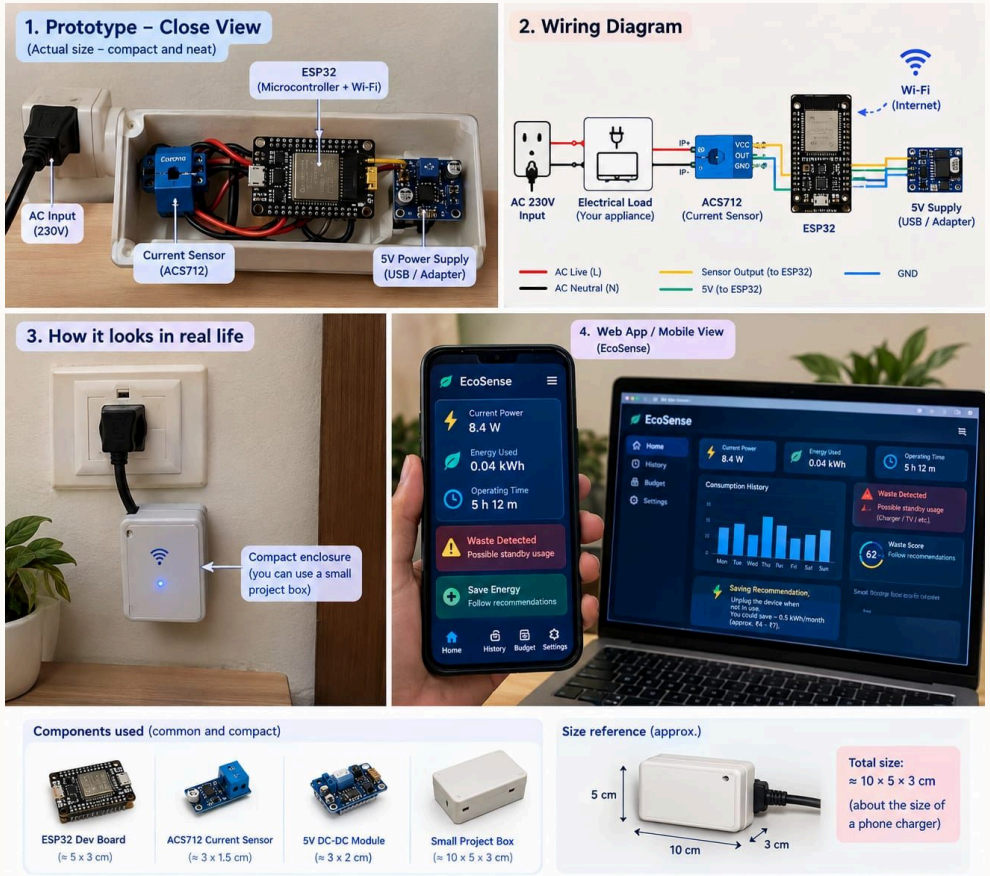
$$P = V \times I$$

Power is voltage multiplied by current.

Energy

$$Energy(kWh) = P(W) \times time(h)/1000$$

Energy depends on power and operating time.



Prototype and system wiring — supplied project image.

Smart Energy Waste Detection

34 / 100

Waste Score

Potential energy waste

NORMAL VS OBSERVED FAN USAGE

Normal Fan Usage

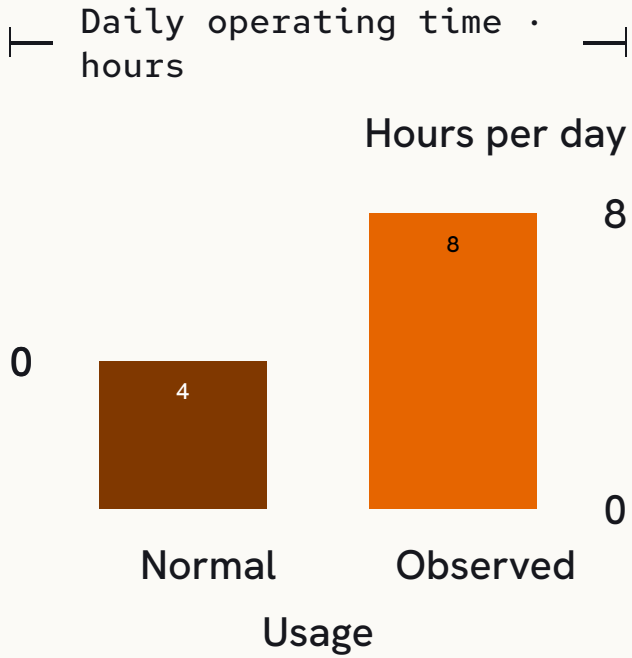
Power: 50 W

Normal time: 4 h/day

Observed Usage

Operating time: 8 h/day

Potential unnecessary usage detected.



WHAT-IF ENERGY SAVER

The user adjusts expected operating time and instantly sees the savings.

Don’t just tell users they are wasting energy — show them how much they can save.

Fan example · 120 W load · ≈ ₹8/kWh

Current usage
6 h/day

Proposed usage
4 h/day

Usage slider (0–12 h): **4 h/day**

Saving	Amount
Energy saved	0.24 kWh/day
Monthly saving	7.2 kWh/month
Money saved	₹58/month

The source example compares 6 h/day with 4 h/day.

ECOSENSE WEB APPLICATION

Illustrative project interface · four views

EcoSense

Illustrative interface mockup · values sourced from the live public demo

Open live EcoSense demo

Live Monitoring · Daily Energy Summary · Smart Alert · Consumption Prediction · Device-Level Breakdown · Waste Score · Recommendation Engine · What-if Energy Saver · Usage Trend · Energy Saving Challenge

Current Power	Daily Energy	Budget
68 W	**7.4 kWh** +42%	**74%** utilized
Monthly Cost	**Prediction**	**Usage Trend**
₹1,920	**12.1 kWh** vs 10	Observed vs normal

Smart Alert · Test Load is operating longer than normal

Device-Level Breakdown

Fan 52 W · 4 h/day normal · 3.2 h observed **Bulb** 18 W · 5 h/day · 4.1 h **Test Load** 85 W · 2 h/day · 6 h

Waste Score · Recommendation Engine

34/100 · Reduce Test Load toward 2 h/day

What-if Saver · Energy Saving Challenge

0.12 kWh/day saved · 3.60 kWh/month · ₹29 · 33%

1. **Dashboard**
   Power, energy, budget, cost, alert
2. **Load Monitoring**
   Power, duration, normal usage
3. **Waste Detective**
   Waste Score and action
4. **What-If Saver**
   Usage, energy, money saved

IMPACT & FUTURE SCOPE

Impact

- Makes electricity consumption visible
- Detects unusual usage patterns
- Helps identify potential energy waste
- Encourages efficient energy use
- Shows possible energy and cost savings
- Supports informed energy decisions

Future scope

- AI-based appliance recognition
- Multiple appliance monitoring
- Automatic switching / control
- Solar energy integration
- Smart-home integration
- Advanced fault detection
- Personalized recommendations

From measuring energy to making energy-saving decisions.

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ECOSENSE 2.0 · ENGINEERING OVERVIEW

ADVANTAGES & LIMITATIONS

Smart Energy Monitoring, Waste Detection & Saving Recommendation System

ADVANTAGES

- Real-time energy monitoring
- Detects unusual energy consumption
- Identifies potential energy waste
- Provides energy-saving recommendations
- Estimates potential energy and cost savings
- Scalable for homes, laboratories, classrooms and offices

LIMITATIONS & FUTURE IMPROVEMENTS

- Initial hardware setup is required
- Current prototype monitors a limited number of loads
- Sensor accuracy depends on calibration
- Electricity cost is an estimate
- Internet connection is required for web-based monitoring
- Multiple sensors can be added for detailed appliance-level monitoring

ECOSENSE 2.0 · PROJECT CLOSING

THANK YOU

EcoSense 2.0 – Smart Energy Monitoring, Waste Detection & Saving Recommendation System

Team Kinetic Minds

Department of Physics, Rathinam Global Deemed to be University

Questions?